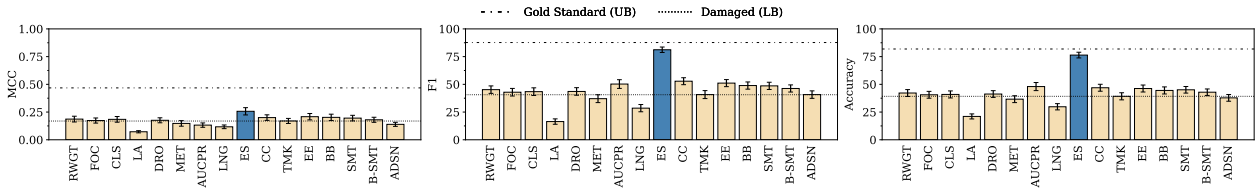
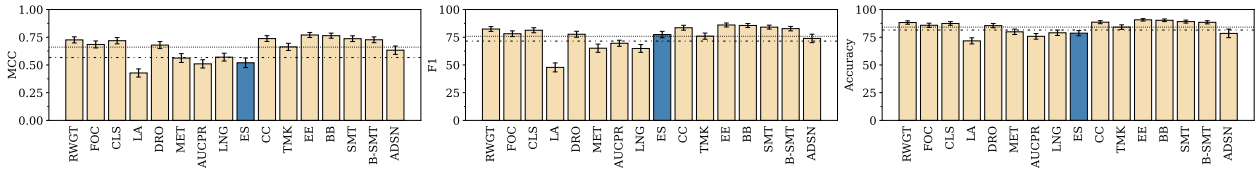
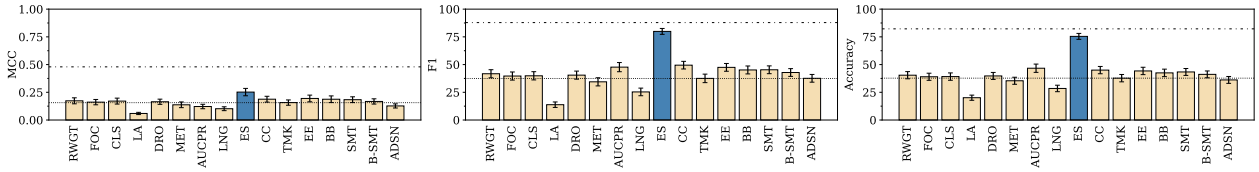
Rebuttal Figures and Tables



(a) Overall performance under LMM imbalance



(b) Performance on likely low-LMM regions



(c) Performance on severe-LMM regions

Figure 1 Average performance under LMM imbalance, across all base classifiers and datasets. Overall performance is further disaggregated into low-LMM and severe-LMM regions.

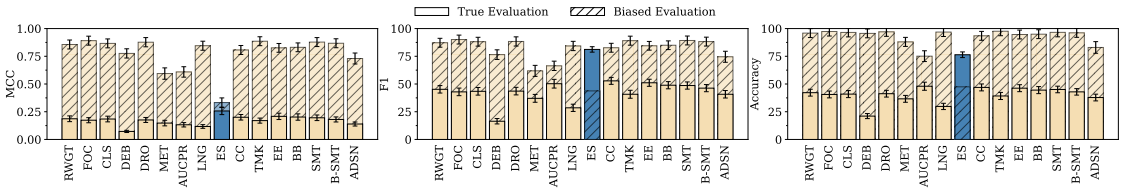


Figure 2 Average performance from hold-out sets sampled from LMM data, across all base classifiers and datasets. (Please note original submission mistakenly included only results for logistic regression base classifier. We sincerely apologies for the error. The new results are more favorable for our claim.)

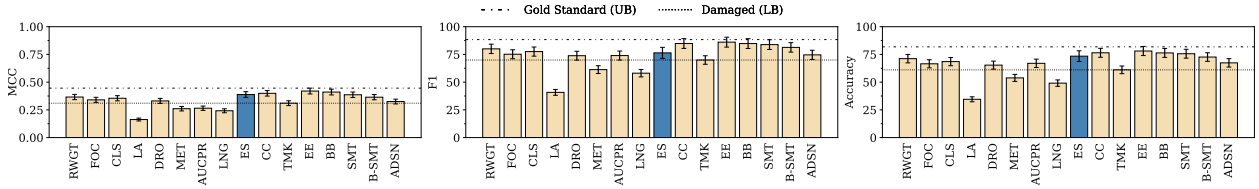


Figure 3 **Performance under class-count imbalance.**

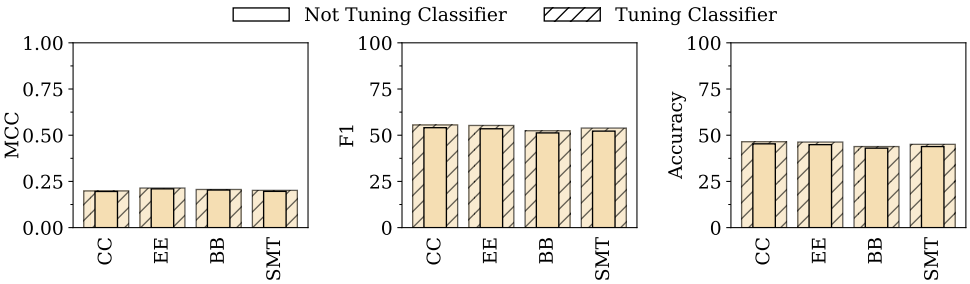


Figure 4 **Tuning base classifiers.**

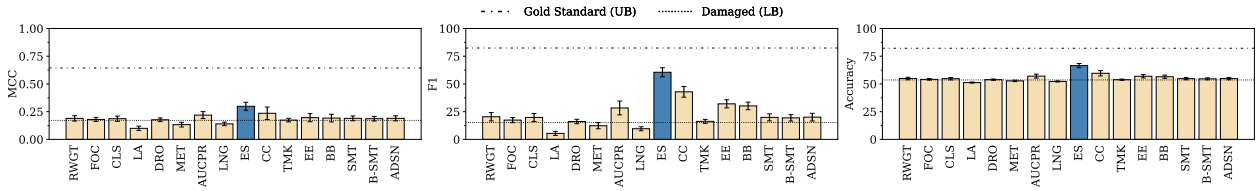


Figure 5 **Results on new dataset WBank.**

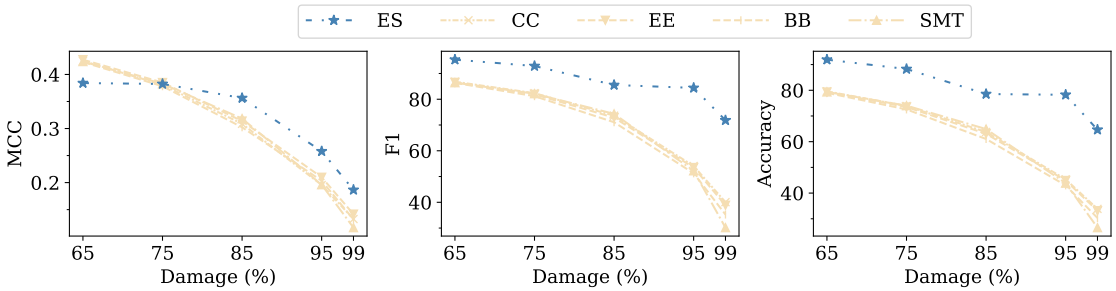
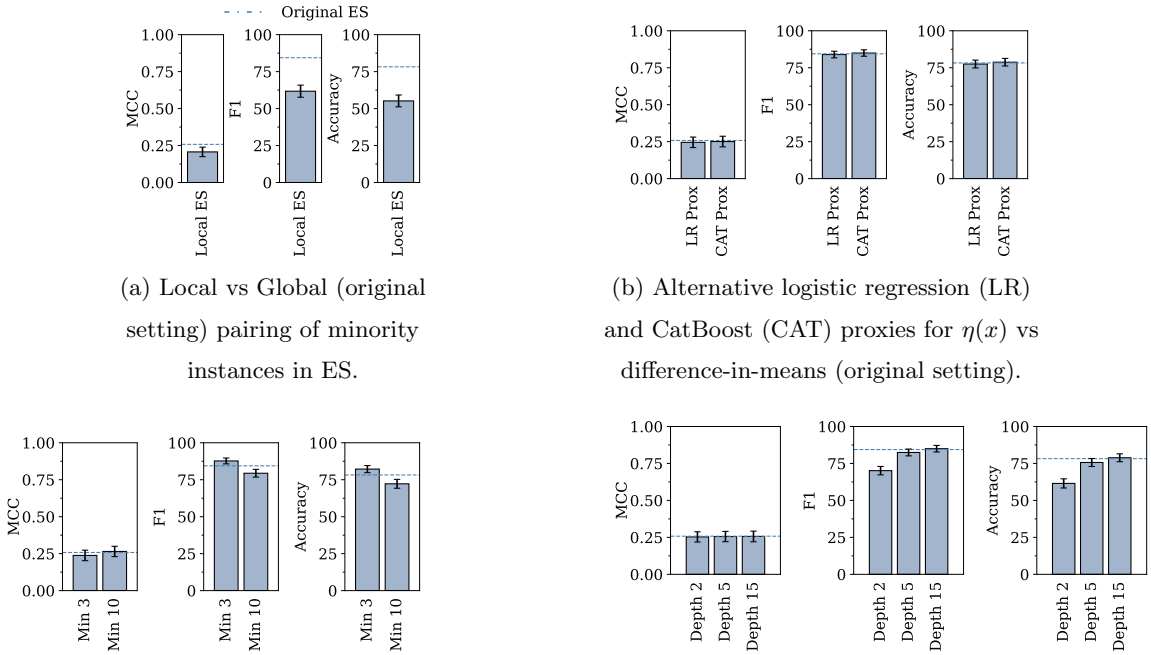


Figure 6 **Results under varying LMM imbalance.**

Table 1 **Balanced datasets, their number of instances, and dimension of X. New WBank dataset is adopted from Field et al. (2021)**

Dataset	N	d	Dataset	N	d
page-blocks0	7860	10	segment0	3166	19
abalone19	6626	8	shuttle-c0-vs-c4	2728	9
yeast3	2114	8	yeast4	2294	8
yeast5	2304	8	yeast6	2318	8
WBank	8065	48			



(a) Local vs Global (original setting) pairing of minority instances in ES.

(b) Alternative logistic regression (LR) and CatBoost (CAT) proxies for $\eta(x)$ vs difference-in-means (original setting).

(c) Effects of minimum leaf size constraint on ES (original has a constraint of 5)

(d) Effects of maximum depth constraint on ES (original has no limit)

Figure 7 Ablation Studies.

Table 2 Per-classifier results under LMM imbalance.

Metric	Classifier	UB	Cost-based					Debias	Ensemble-based			Under-sampling		Over-sampling			Optimization		LB
		GLD	RWGT	FOC	CLS	MET	LNG	LA	ES	EE	BB	CC	TMK	SMT	B-SMT	ADSN	DRO	AUCPR	DMG
MCC	LR	0.29	0.13	0.10	0.12	0.07	0.06	0.04	**0.24**	0.14	0.14	0.13	0.08	0.13	0.11	0.09	0.10	0.16	0.08
	SVM	0.33	0.12	0.13	0.12	0.11	0.10	0.08	**0.16**	**0.16**	**0.16**	**0.16**	0.13	0.15	0.14	0.11	0.13	0.08	0.13
	RF	0.65	0.24	0.25	0.24	0.29	0.18	0.10	**0.36**	0.31	0.29	0.27	0.29	0.28	0.26	0.18	0.27	0.18	0.29
	NN	0.31	0.16	0.14	0.15	0.11	0.09	0.06	**0.21**	0.17	0.16	0.16	0.13	0.15	0.15	0.11	0.13	0.10	0.12
	CAT	0.65	0.29	0.25	0.28	0.17	0.14	0.07	**0.31**	0.27	0.27	0.26	0.21	0.28	0.24	0.17	0.24	0.08	0.22
	Avg	0.45	0.19	0.17	0.18	0.15	0.11	0.07	**0.26**	0.21	0.20	0.20	0.17	0.20	0.18	0.13	0.17	0.12	0.17
F1	LR	0.77	0.43	0.31	0.39	0.31	0.14	0.08	**0.72**	0.44	0.43	0.43	0.22	0.41	0.38	0.36	0.31	0.49	0.22
	SVM	0.86	0.38	0.50	0.35	0.36	0.38	0.28	**0.82**	0.52	0.49	0.50	0.49	0.49	0.45	0.41	0.50	0.31	0.49
	RF	0.99	0.53	0.54	0.54	0.65	0.42	0.24	**0.92**	0.64	0.60	0.66	0.63	0.64	0.60	0.51	0.59	0.42	0.63
	NN	0.82	0.50	0.40	0.47	0.40	0.26	0.13	**0.86**	0.52	0.50	0.51	0.37	0.49	0.48	0.40	0.40	0.58	0.37
	CAT	0.98	0.58	0.56	0.57	0.29	0.35	0.15	**0.89**	0.56	0.54	0.59	0.47	0.59	0.57	0.48	0.55	0.85	0.47
	Avg	0.88	0.48	0.46	0.46	0.40	0.31	0.18	**0.84**	0.54	0.51	0.54	0.44	0.52	0.50	0.43	0.47	0.53	0.44
Accuracy	LR	0.65	0.34	0.25	0.31	0.25	0.14	0.10	**0.63**	0.35	0.34	0.35	0.19	0.33	0.30	0.28	0.26	0.42	0.19
	SVM	0.77	0.31	0.40	0.28	0.30	0.31	0.24	**0.75**	0.41	0.39	0.41	0.40	0.39	0.36	0.32	0.40	0.26	0.40
	RF	0.98	0.47	0.48	0.48	0.57	0.38	0.22	**0.89**	0.57	0.53	0.58	0.56	0.57	0.53	0.45	0.52	0.37	0.56
	NN	0.72	0.39	0.32	0.37	0.32	0.22	0.14	**0.80**	0.42	0.40	0.41	0.30	0.39	0.38	0.32	0.32	0.50	0.30
	CAT	0.97	0.51	0.49	0.51	0.29	0.31	0.16	**0.85**	0.49	0.48	0.52	0.42	0.52	0.50	0.42	0.48	0.78	0.42
	Avg	0.82	0.40	0.39	0.39	0.35	0.27	0.17	**0.78**	0.45	0.43	0.45	0.37	0.44	0.41	0.36	0.40	0.47	0.37

References

Field E, Pande R, Rigol N, Schaner S, Troyer Moore C (2021) On her own account: How strengthening women's financial control impacts labor supply and gender norms. *American Economic Review* 111(7):2342–2375.